

STAT 192 Capstone Project

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Empathy in Scientific Communication

Analysis of Pre-Survey Data

To begin our analysis of the pre-survey data collected during our experiment, we must first properly setup, clean, and pre-process the data into a useable format for calculating statistics and creating models.

We first clean the data by dropping any columns that will not be useful for future models or calculating statistics. We'll then make two copies of this survey dataframe, each isolating the Jefferson and Toronto scales repectively.

Our next goal is to convert the survey answers into numerical format, so we can calculate score statistics. Usually, we want to avoid encoding our categorical features using a numeric key based system, because that implies a relationship between and/or across variables. However, in this case, since the survey properly follows the guidelines established by each scale, this linear relationship is built into the survey structure. Therefore, we can convert our categorical answers to a number which we can then use to calculate statistics.

The **Jefferson Scale** is a 7-point likert agreement based scale (Extremely Disagree -> Extremely Agree). While the original **Toronto Scale** is a 5-point likert oftenality based scale. (Never -> Often).

These values are mapped linearly to the following outcome score:

Survey Label	Score	Reverse Score	
Extremely Disagree	1	7	
Disagree	2	6	
Somewhat Disagree	3	5	
Neutral	4	4	
Somewhat Agree	5	3	
Agree	6	2	
Extremely Agree	7	1	

Survey Label	Score	Reverse Score	Corresponding Toronto Label
Disagree	0	4	Never
Somewhat Disagree	1	3	Rarely
Neutral	2	2	Sometimes
Somewhat Agree	3	1	Often

Survey Label	Score	Reverse Score	Corresponding Toronto Label
Agree	4	0	Always

To maintain consistency in the survey methodology, structure, and question wording. The Toronto scale was modified and mapped to 5 levels of the agreement based scale. In order to retain the meaning of the Toronto scale, the questions had to be modified to allow the questions to make sense. Due to this modification of the scale, we are *unable to compare across our two scales* and we must treat the two scales independently.

We also need to remove DID # 40, 43, 81, 102, 107, 125, 126, 144, and 149.

The cleaned data set, `pre_survey_clean`, for all student's pre-survey is attached.

`pre_survey_clean` has the following features:

- `DID #` - ID Number (Double)
- `Survey` - Pre/Post (Factor)
- `First Generation Status` - Yes/No (Factor)
- `Gender` - Man/Woman/Other (Factor)
- `Transfer Student` - Yes/No (Factor)
- `Upper v Lower` - U/L (Factor)
- `Race/Ethnicity` - Race or Ethnicity (Factor)
- `Class Level` - Freshman/Sophomore/Junior/Senior (Factor)
- `J_total` - Jefferson scale score total (Double)
- `T_total` - Toronto scale score total (Double)
- `Major` - Major Title (Char)
- `STEM` - STEM/non-STEM (Factor)

Checking our Assumptions

Verifying Scale Modification

I first want to ensure the meaning of the modified Toronto scale still holds. To check if this is true, we can check the correlation matrix between our Jefferson scale scores and Toronto scale scores. Since they are both measuring empathy levels in scientific communication, they should *at least* be moderately to highly correlated.

```
JT_data <- pre_survey_clean[, c("J_total", "T_total")]
cor(JT_data, use = "pairwise.complete.obs")
```

Which gives us a correlation matrix of:

J_total	T_total
1.0000000	0.5935951
0.5935951	1.0000000

We can see there is correlation between the score totals, therefore we can declare our modified Toronto scale still measures empathy levels.

We also want to check our assumptions for future modeling and testing. Let's check our a summary of the basic statistics of our data to see if we find anything worth exploring further.

Exploratory Data Analysis

```
summary(pre_survey_clean)
```

Variable	Values / Statistics
J_total	Min: 54, 1st Q: 69.25, Median: 76, Mean: 75.76, 3rd Q: 82, Max: 95
T_total	Min: 30, 1st Q: 44, Median: 48.5, Mean: 47.88, 3rd Q: 52.75, Max: 60
Variable	Values / Statistics
DID #	Min: 1, 1st Q: 58.25, Median: 116.50, Mean: 116.57, 3rd Q: 175.75, Max: 231
Survey	Pre: 222, Post: 0
First Generation Status	No: 158, Yes: 63, NA: 1
Gender	Woman: 175, Man: 45, Other: 1, NA: 1
Transfer Student	No: 134, Yes: 87, NA: 1
Upper vs Lower	Lower: 110, Upper: 112
Race/Ethnicity	Asian: 66, Hispanic/Latino: 65, White: 48, Two or More: 25, African American: 9, Other: 5, NA: 4
Class Level	Freshman: 5, Sophomore: 45, Junior: 57, Senior: 113, NA: 2
Major	Character variable (n = 222)
STEM	STEM: 100, Non-STEM: 122

We can see our pre-survey data looks good. We have a goood representative sample of 103 STEM majors and 128 non-STEM majors.

We can investigate a little further by grouping the STEM and non-STEM categories then exploring the averages for each scale.

```
pre_survey_clean %>%
  group_by(STEM) %>%
  summarise(
    n = n(),
    mean_J = mean(J_total, na.rm = TRUE),
    sd_J = sd(J_total, na.rm = TRUE),
    mean_T = mean(T_total, na.rm = TRUE),
    sd_T = sd(T_total, na.rm = TRUE)
  )
```

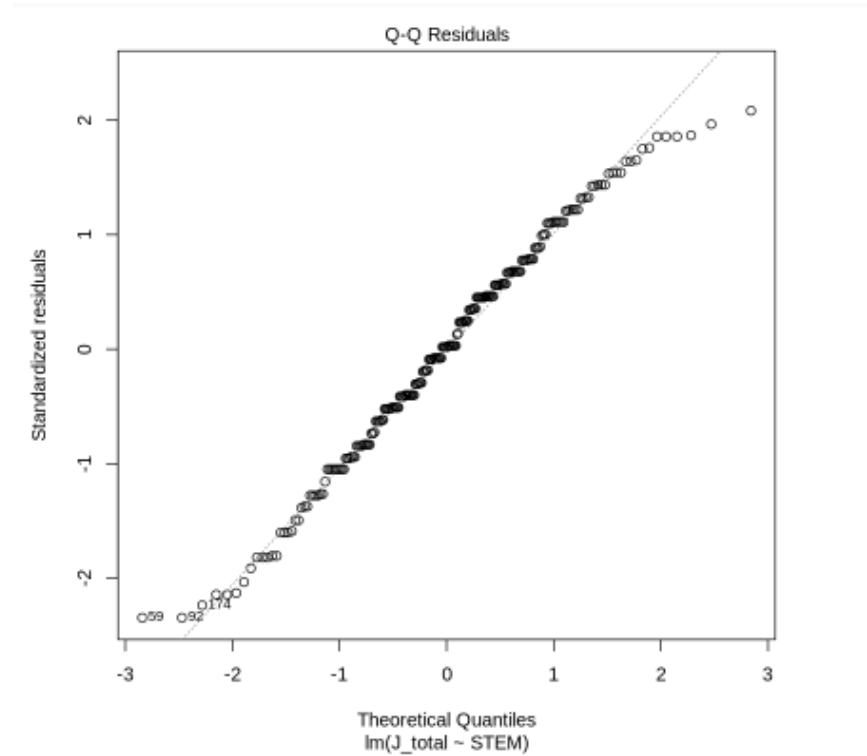
STEM	n	mean_J	sd_J	mean_T	sd_T
STEM	100	75.82000	9.494688	47.93000	6.115480
Non-STEM	122	75.71311	9.140302	47.83607	5.738604

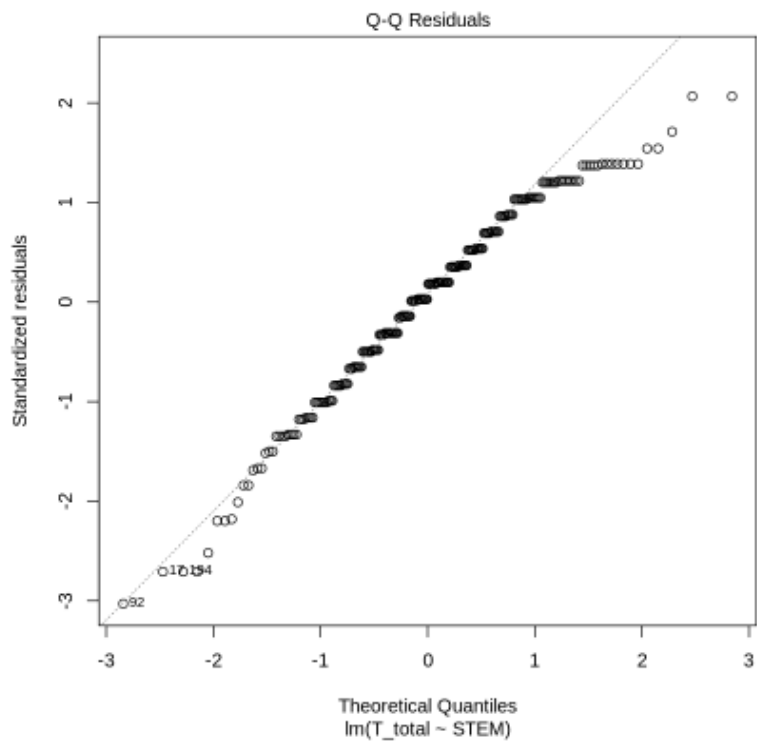
Residual vs Fitted and Q-Q Plots

```
J_model <- lm(J_total ~ STEM, data = pre_survey_clean)
plot(J_model, which = 1)

T_model <- lm(T_total ~ STEM, data = pre_survey_clean)
plot(T_model, which = 1)

plot(J_model, which = 2)
plot(T_model, which = 2)
```





Analysis

Let's begin our analysis of the pre-survey data by conducting some independent sample t-tests to find the average score of STEM vs non-STEM students for both the Jefferson and Toronto scales. Since individual Likert items are ordinal, we report Mann–Whitney tests as a nonparametric comparison. However, because total scores represent the sum of many Likert items, they approximate normality (CLT), so we also report Welch’s t-tests for robustness.

Wilcoxon Rank Sum (Assuming non-normal data)

We want to compare the differences between STEM and non-STEM average scores for each scale. We will need to do these analyses independently, since we can't compare across scales.

```
wilcox.test(J_total ~ STEM, data = pre_survey_clean)
wilcox.test(T_total ~ STEM, data = pre_survey_clean)
```

Scale	Test	W Statistic	p-value
Jefferson scale	Wilcoxon rank sum (STEM vs non-STEM)	6141	0.9322
Toronto scale	Wilcoxon rank sum (STEM vs non-STEM)	6254	0.7468

Large p-value indicates no evidence of a difference in the distribution of Jefferson empathy scores between STEM and non-STEM students. Similarly, large p-value indicates no statistically significant difference in the distribution of Toronto empathy scores.

Therefore we can conclude that there is no evidence that academic major type influences empathy scores.

Welch's t-test

We can also conduct a Welch's t-test to check for robustness in our model. This is allowed since we summed up the scores to create a total score column. These summed scores behave more like continuous data due to the central limit theorem, making the Welch's t-test acceptable even if individual items were ordinal. Welch's t-test is more robust than the standard t-test because it does not assume equal variances between groups.

```
t.test(J_total ~ STEM, data = pre_survey_clean, var.equal = FALSE)
t.test(T_total ~ STEM, data = pre_survey_clean, var.equal = FALSE)
```

Scale	Test	t-statistic	df	p-value	95% CI (lower)	95% CI (upper)	STEM mean	Non-STEM mean
Jefferson scale	Welch Two Sample t-test	0.084865	208.23	0.9325	-2.376079	2.589850	75.82000	75.71311
Toronto scale	Welch Two Sample t-test	0.11706	205.78	0.9069	-1.488138	1.676007	47.93000	47.83607

As we can see, the average score for STEM students in the pre-survey is \$75.82\$ and non-STEM students is \$75.71\$ on the Jefferson scale. The average score for STEM students in the pre-survey is \$47.93\$ and non-STEM students is \$47.84\$ on the Toronto scale.

Once again, we see large p-values which means there is no statistically significant difference in the means across groups.

Further Analysis

First Generation Status

For fun, lets look at some relationships between other variables and total scores for each scale in the pre-survey results.

Let's first investigate **First Generation Status** to see if there is a difference in average scores between first generation students and others.

```
wilcox.test(J_total ~ `First Generation Status`, data = pre_survey_clean)
wilcox.test(T_total ~ `First Generation Status`, data = pre_survey_clean)
```

Scale	Test	W Statistic	p-value
Jefferson scale	Wilcoxon rank sum (1st Gen vs non-1st Gen)	5090	0.7931
Toronto scale	Wilcoxon rank sum (1st Gen vs non-1st Gen)	4682.5	0.4926

Large p-value indicates no evidence of a difference in the distribution of Jefferson empathy scores between 1st gen and non-1st gen students. Similarly, large p-value indicates no statistically significant difference in the distribution of Toronto empathy scores.

Therefore we can conclude that there is no evidence that first generation status influences empathy scores.

```
t.test(J_total ~ `First Generation Status`, data = pre_survey_clean, var.equal = FALSE)
t.test(T_total ~ `First Generation Status`, data = pre_survey_clean, var.equal = FALSE)
```

Scale	Test	t-statistic	df	p-value	95% CI (lower)	95% CI (upper)	Non-1st Gen mean	1st Gen mean
Jefferson scale	Welch Two Sample t-test	0.26257	95.47	0.7934	-2.626341	3.427024	75.89241	75.49206
Toronto scale	Welch Two Sample t-test	-0.45451	98.42	0.6505	-2.313171	1.451005	47.79114	48.22222

Similar to before, the means for each group are almost equivalent, suggesting that there is no significant difference in means between 1st gen and non-1st gen students for both scales. We also see large p-values which means there is no statistically significant difference in the means across groups.

Upper v Lower

We'll also investigate **Upper v Lower** to see if there is a difference in average scores between upperclassmen and lowerclassmen.

```
wilcox.test(J_total ~ `Upper v Lower`, data = pre_survey_clean)
wilcox.test(T_total ~ `Upper v Lower`, data = pre_survey_clean)
```

Scale	Test	W Statistic	p-value
Jefferson scale	Wilcoxon rank sum (Upperclassmen vs Lowerclassmen)	5901	0.5888
Toronto scale	Wilcoxon rank sum (Upperclassmen vs Lowerclassmen)	6104.5	0.9084

Large p-value indicates no evidence of a difference in the distribution of Jefferson empathy scores between upperclassmen and lowerclassmen students. Similarly, large p-value indicates no statistically significant difference in the distribution of Toronto empathy scores.

Therefore we can conclude that there is no evidence that class level influences empathy scores.

```
t.test(J_total ~ `Upper v Lower`, data = pre_survey_clean, var.equal = FALSE)
t.test(T_total ~ `Upper v Lower`, data = pre_survey_clean, var.equal = FALSE)
```

Scale	Test	t-statistic	df	p-value	95% CI (lower)	95% CI (upper)	Lowerclassmen mean	Upperclassmen mean
Jefferson scale	Welch Two Sample t-test	-0.48744	219.56	0.6264	-3.066040	1.850131	75.45455	76.06250

Scale	Test	t-statistic	df	p-value	95% CI (lower)	95% CI (upper)	Lowerclassmen mean	Upperclassmen mean
Toronto scale	Welch Two Sample t-test	0.076753	219.76	0.9389	-1.502267	1.624020	47.90909	47.84821

Similar to before, the means for each group are almost equivalent, suggesting that there is no significant difference in means between upperclassmen and lowerclassmen students for both scales. Lowerclassmen actually score an average of 1 point higher than upperclassmen on the Jefferson scale. We also see large p-values which means there is no statistically significant difference in the means across groups.

Transfer Student

We'll also investigate **Transfer Student** to see if there is a difference in average scores between transfers and non-transfers.

```
wilcox.test(J_total ~ `Transfer Student`, data = pre_survey_clean)
wilcox.test(T_total ~ `Transfer Student`, data = pre_survey_clean)
```

Scale	Test	W Statistic	p-value
Jefferson scale	Wilcoxon rank sum (Yes vs No)	5714.5	0.806
Toronto scale	Wilcoxon rank sum (Yes vs No)	5896.5	0.8851

Large p-value indicates no evidence of a difference in the distribution of Jefferson empathy scores between transfer and non-transfer students. Similarly, large p-value indicates no statistically significant difference in the distribution of Toronto empathy scores.

Therefore we can again conclude that there is no evidence that transfer status influences empathy scores.

```
t.test(J_total ~ `Transfer Student`, data = pre_survey_clean, var.equal = FALSE)
t.test(T_total ~ `Transfer Student`, data = pre_survey_clean, var.equal = FALSE)
```

Scale	Test	t-statistic	df	p-value	95% CI (lower)	95% CI (upper)	Non-Transfer mean	Transfer mean
Jefferson scale	Welch Two Sample t-test	-0.31128	177.21	0.7560	-2.962252	2.155081	75.61940	76.02299
Toronto scale	Welch Two Sample t-test	0.082692	187.91	0.9342	-1.525271	1.658741	47.94030	47.87356

Similar to before, the means for each group are almost equivalent, suggesting that there is no significant difference in means between transfer and non-transfer students for both scales. We also again see large p-values which means there is no statistically significant difference in the means across groups.

Gender

We'll lastly investigate **Gender** to see if there is a difference in average scores between men and women. We have to remove 2 rows from our data set that responded "Other" or NA since we can only run our analysis on two groups.

```
wilcox.test(J_total ~ `Gender`, data = pre_survey_clean)
wilcox.test(T_total ~ `Gender`, data = pre_survey_clean)
```

Scale	Test	W Statistic	p-value
Jefferson scale	Wilcoxon rank sum (Man vs Woman)	2907.5	0.006826
Toronto scale	Wilcoxon rank sum (Man vs Woman)	2598	0.0004293

There is a statistically significant difference in J-scale total scores between gender groups. Because the p-value is < .01, we reject the null hypothesis that the two gender groups have the same distribution.

There is a highly significant difference in T-scale total scores between gender groups. With a p-value well below .001, we again reject the null hypothesis.

This indicates that gender groups differ in their T_total empathy scores, and the difference is large enough that it is unlikely due to random variation.

```
t.test(J_total ~ `Gender`, data = pre_survey_clean, var.equal = FALSE)
t.test(T_total ~ `Gender`, data = pre_survey_clean, var.equal = FALSE)
```

Scale	Test	t-statistic	df	p-value	95% CI (lower)	95% CI (upper)	Man mean	Women mean
Jefferson scale	Welch Two Sample t-test	-2.6827	67.522	0.009174	-7.258039	-1.065771	72.46667	76.62857

Toronto scale	Welch Two Sample t-test	-3.4144	58.571	0.001166	-5.969893	-1.557726	44.93333	48.69714
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There is a statistically significant difference in J_total scores between men and women. Because the p-value is < .01, we reject the null hypothesis that the two groups have equal means. Women scored significantly higher on the J-scale than men. The estimated average difference between men and women is about 4.16 points.

There is a highly significant difference in T_total scores between men and women. With a p-value around .001, we again reject the null hypothesis. Women again scored significantly higher than men. The estimated mean difference between men and women is about 3.76 points.

Therefore we can finally conclude that gender is strongly associated with differences in T-scale scores, with women reporting higher levels of empathy on the Toronto scale.

Visualizations

Below are some box and whisker plots for each scale to visualize the average scores of STEM vs non-STEM students taking the pre-survey. I have hidden them behind a spoiler section so you can generate your own plots first, then

compare your results to ours.

► Example box plots for STEM vs non-STEM students

